Air Quality API 0.1.0 OAS 3.0

/api/schema/

Coursework API for city-level air quality data.

analytics

GET /api/analytics/city-trend/

获取指定城市某污染物的时间趋势数据 参数: city - 城市名 pollutant - 污染物类型 (pm25/pm10/no2/co/aqi) 返回: 按日期排序的污染物数值列表

Parameters

Cancel

Name	Description
city * required string *(query)*	城市名称（精确匹配，不区分大小写） cd
pollutant * required string *(query)*	污染物类型 aqi

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/analytics/city-trend/?city=cd&pollutant=aqi' \
  -H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/api/analytics/city-trend/?city=cd&pollutant=aqi
```

Server response

Code	Details
200	

Response body


```
{
  "city": "cd",
  "pollutant": "aqi",
  "count": 1,
  "data": [
    {
      "date": "2026-04-19",
      "value": 234
    }
  ]
}
```


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Response headers

```
allow: GET,HEAD,OPTIONS
content-length: 84
content-type: application/json
cross-origin-opener-policy: same-origin
date: Sun,19 Apr 2026 18:06:15 GMT
referrer-policy: same-origin
server: WSGIServer/0.2 CPython/3.13.13
vary: Accept
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header.	*No links*

Example Value Schema


```
{
  "city": "string",
  "pollutant": "string",
  "count": 0,
  "data": [
    {
      "date": "2026-04-19",
      "value": 0
    }
  ]
}
```


400	Media type **application/json**	*No links*

Code	Description	Links

	Example Value Schema
	
	{
	"error": "city 参数是必需的"
	}	

404	Media type
	application/json
	
	Example Value Schema
	
	{
	"message": "未找到城市 Delhi 的记录"
	}
		No links

records

GET /api/records/

Parameters Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/records/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/api/records/

Server response

Code	Details
200	**Response body**

Code	Details


```
[
  {
    "id": 30061,
    "city": "Aizawl",
    "date": "2020-07-01",
    "pm25": 4.49,
    "pm10": 5.39,
    "no2": 0.07,
    "co": 0.02,
    "aqi": 20,
    "created_at": "2026-04-19T15:27:39.091720Z",
    "updated_at": "2026-04-19T15:27:39.091735Z"
  },
  {
    "id": 29952,
    "city": "Ahmedabad",
    "date": "2020-06-30",
    "pm25": 40.02,
    "pm10": 121.1,
    "no2": 58.92,
    "co": 0.73,
    "aqi": 98,
    "created_at": "2026-04-19T15:27:38.173971Z",
    "updated_at": "2026-04-19T15:27:38.173986Z"
  },
  {
    "id": 30060,
    "city": "Aizawl",
```


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Response headers

```
allow: GET,POST,HEAD,OPTIONS
content-length: 195552
content-type: application/json
cross-origin-opener-policy: same-origin
date: Sun,19 Apr 2026 17:36:51 GMT
referrer-policy: same-origin
server: WSGIServer/0.2 CPython/3.13.13
vary: Accept
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header.	*No links*

Example Value Schema


```
[
  {
    "id": 0,
    "city": "string",
    "date": "2026-04-19",
    "pm25": 0,
    "pm10": 0,
    "no2": 0,
    "co": 0,
    "aqi": 9223372036854776000,
    "created_at": "2026-04-19T18:09:17.325Z",
    "updated_at": "2026-04-19T18:09:17.325Z"
```

Code	Description	Links

```
    }
  ]
```

POST /api/records/

Parameters Cancel Reset

No parameters

Request body required application/json

Edit Value Schema

```
{
  "city": "cd",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 234
}
```

Execute Clear

Responses

Curl

```
curl -X 'POST' \
  'http://127.0.0.1:8000/api/records/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -H 'X-CSRFTOKEN: nNQtMpB3EnOFoxOibrAadEB0nrjulnfsvITvoWUa54nu9sAmefG4uh6ncoNekgcN' \
  -d '{
  "city": "cd",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
```

```
  "aqi": 234
}'
```

Request URL

```
http://127.0.0.1:8000/api/records/
```

Server response

Code	Details
201	**Response body**

```
{
  "id": 30557,
  "city": "cd",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 234,
  "created_at": "2026-04-19T17:37:21.809684Z",
  "updated_at": "2026-04-19T17:37:21.809704Z"
}
```

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Response headers

```
allow: GET,POST,HEAD,OPTIONS
content-length: 181
content-type: application/json
cross-origin-opener-policy: same-origin
date: Sun,19 Apr 2026 17:37:21 GMT
referrer-policy: same-origin
server: WSGIServer/0.2 CPython/3.13.13
vary: Accept
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
201	Media type **application/json** Controls `Accept` header.	*No links*

Example Value Schema

```
{
  "id": 0,
  "city": "string",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 9223372036854776000,
  "created_at": "2026-04-19T18:09:17.326Z",
  "updated_at": "2026-04-19T18:09:17.326Z"
}
```

GET /api/records/{id}/

Parameters

Cancel

Name	Description
id * required integer *(path)*	4

Execute | Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/api/records/4/' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8000/api/records/4/

Server response

Code	Details
200	

Response body


```
{
  "id": 4,
  "city": "Ahmedabad",
  "date": "2019-05-18",
  "pm25": 34.11,
  "pm10": 138.31,
  "no2": 75.23,
  "co": 13.27,
  "aqi": 359,
  "created_at": "2026-04-19T17:47:24.591983Z",
  "updated_at": "2026-04-19T17:47:24.591992Z"
}
```


Download

Response headers

```
allow: GET,PUT,PATCH,DELETE,HEAD,OPTIONS
content-length: 193
content-type: application/json
cross-origin-opener-policy: same-origin
date: Sun,19 Apr 2026 18:07:00 GMT
referrer-policy: same-origin
server: WSGIServer/0.2 CPython/3.13.13
vary: Accept
```

Code	Details
	x-content-type-options: nosniff x-frame-options: DENY

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. **Example Value** Schema	*No links*

```
{
  "id": 0,
  "city": "string",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 9223372036854776000,
  "created_at": "2026-04-19T18:09:17.328Z",
  "updated_at": "2026-04-19T18:09:17.328Z"
}
```

PUT /api/records/{id}/

Parameters Cancel Reset

Name	Description
id * required integer *(path)*	5

Request body required application/json

Edit Value Schema

```
{
  "city": "string",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 3445
}
```

Execute | Clear

Responses

Curl

```
curl -X 'PUT' \
  'http://127.0.0.1:8000/api/records/5/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -H 'X-CSRFTOKEN: nNQtMpB3EnOFoxOibrAadEB0nrjulnfsvITvoWUa54nu9sAmefG4uh6ncoNekgcN' \
  -d '{
  "city": "string",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 3445
}'
```

Request URL

http://127.0.0.1:8000/api/records/5/

Server response

Code	Details
200	**Response body**

```
{
  "id": 5,
  "city": "string",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 3445,
  "created_at": "2026-04-19T17:47:24.602364Z",
```

Download

Code	Details


```
  "updated_at": "2026-04-19T18:07:23.518848Z"
}
```


Response headers

```
allow: GET,PUT,PATCH,DELETE,HEAD,OPTIONS
content-length: 182
content-type: application/json
cross-origin-opener-policy: same-origin
date: Sun,19 Apr 2026 18:07:23 GMT
referrer-policy: same-origin
server: WSGIServer/0.2 CPython/3.13.13
vary: Accept
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. **Example Value** Schema	*No links*


```
{
  "id": 0,
  "city": "string",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 9223372036854776000,
  "created_at": "2026-04-19T18:09:17.329Z",
  "updated_at": "2026-04-19T18:09:17.329Z"
}
```


PATCH /api/records/{id}/

Parameters Cancel Reset

Name	Description
id * required integer (path)	4

Request body application/json

Edit Value Schema

```
{
  "pm25": 0
}
```

Execute | Clear

Responses

Curl

```
curl -X 'PATCH' \
  'http://127.0.0.1:8000/api/records/4/' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -H 'X-CSRFTOKEN: nNQtMpB3EnOFoxOibrAadEB0nrjulnfsvITvoWUa54nu9sAmefG4uh6ncoNekgcN' \
  -d '{
  "pm25": 0
}'
```

Request URL

http://127.0.0.1:8000/api/records/4/

Server response

Code	Details
200	**Response body**


```
{
  "id": 4,
  "city": "Ahmedabad",
  "date": "2019-05-18",
  "pm25": 0,
  "pm10": 138.31,
  "no2": 75.23,
  "co": 13.27,
  "aqi": 359,
  "created_at": "2026-04-19T17:47:24.591983Z",
```

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Code	Details


```
  "updated_at": "2026-04-19T18:08:51.576438Z"
}
```


Response headers

```
allow: GET,PUT,PATCH,DELETE,HEAD,OPTIONS
content-length: 191
content-type: application/json
cross-origin-opener-policy: same-origin
date: Sun,19 Apr 2026 18:08:51 GMT
referrer-policy: same-origin
server: WSGIServer/0.2 CPython/3.13.13
vary: Accept
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*


```
{
  "id": 0,
  "city": "string",
  "date": "2026-04-19",
  "pm25": 0,
  "pm10": 0,
  "no2": 0,
  "co": 0,
  "aqi": 9223372036854776000,
  "created_at": "2026-04-19T18:09:17.330Z",
  "updated_at": "2026-04-19T18:09:17.330Z"
}
```


DELETE /api/records/{id}/

Parameters Cancel

Name	Description
id * required integer *(path)*	4

Execute | Clear

Responses

Curl

```
curl -X 'DELETE' \
  'http://127.0.0.1:8000/api/records/4/' \
  -H 'accept: */*' \
  -H 'X-CSRFTOKEN: nNQtMpB3EnOFoxOibrAadEB0nrjulnfsvITvoWUa54nu9sAmefG4uh6ncoNekgcN'
```

Request URL

```
http://127.0.0.1:8000/api/records/4/
```

Server response

Code	Details
204	**Response headers** allow: GET,PUT,PATCH,DELETE,HEAD,OPTIONS content-length: 0 cross-origin-opener-policy: same-origin date: Sun,19 Apr 2026 18:09:17 GMT referrer-policy: same-origin server: WSGIServer/0.2 CPython/3.13.13 vary: Accept x-content-type-options: nosniff x-frame-options: DENY

Responses

Code	Description	Links
204	No response body	*No links*

schema

GET /api/schema/

Schemas

AirQualityRecord

CityTrendResponse

PatchedAirQualityRecord

TrendDataItem